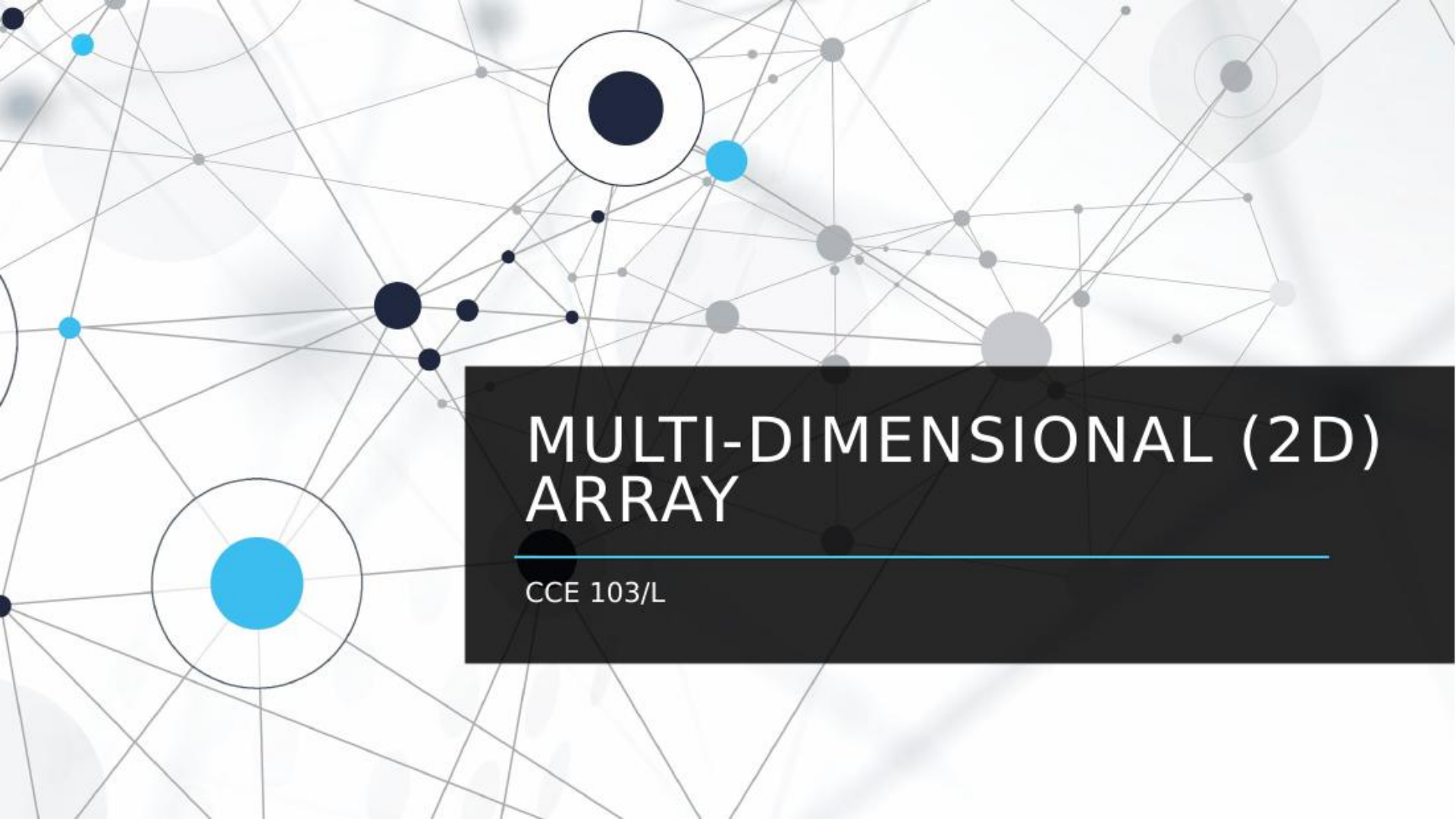




MULTI-DIMENSIONAL (2D) ARRAY

CCE 103/L

OBJECTIVE

- To understand the concept of multi-dimensional array.
- Learn how to declare and initialize multi-dimensional array.
- Write Java program implementing multi-dimensional array.
- Learn the concept of Nested For Loop and to access element of a 2D array.

MULTI-DIMENSIONAL ARRAY (2D ARRAY)

- ❑ An array of arrays. Basically looks like a table with rows and columns.
- ❑ It can only store one datatype at a time.
- ❑ It useful when you want to store data as a tabular form, like a table with rows and columns.

SYNTAX

- Add each array within its own set of **curly brace**.

data_type [][] array_name;

datatype identifier [][] = new datatype [row][cols];

Example:

```
public class Array {  
  
    public static void main(String[] args) {  
  
        int[][] myNumbers = { {1, 2, 3, 4}, {5, 6, 7} };  
        String [][] username = { {"Kate", "Mark", "John"}, {"Kate123", "Mark345", "John456"} };  
    }  
}
```

	Column 1	Column 2	Column 3	Column 4
Row 1	a[0][0]	a[0][1]	a[0][2]	a[0][3]
Row 2	a[1][0]	a[1][1]	a[1][2]	a[1][3]
Row 3	a[2][0]	a[2][1]	a[2][2]	a[2][3]

ACCESSING 2D ARRAY

To access the elements of 2D-Array, specify two indexes:

- One for the array.
- One for the element inside that array.

```
public class Array {  
  
    public static void main(String[] args) {  
  
        int[][] myNumbers = { {1, 2, 3, 4}, {5, 6, 7} };  
        String [][] username = { {"Kate", "Mark", "John"}, {"Kate123", "Mark345", "John456"} };  
        System.out.println(myNumbers[0][2]);  
        System.out.println(username[1][2]);  
    }  
}
```

Output - Array (run)

```
run:  
3  
John456  
BUILD SUCCESSFUL (total time: 0 seconds)
```

ASSIGNING VALUE IN 2D ARRAY

Syntax:

Identifier [row][col] = value;

```
username[0][2]="RJ";  
System.out.println(username[0][2]);
```

Output - Array (run)



```
run:  
RJ  
BUILD SUCCESSFUL (total time: 0 seconds)  
|
```

NESTED FOR LOOP

A for loop inside another for loop.

```
public static void main(String[] args) {  
  
    for (int i = 1; i <= 2; i++) {  
        System.out.println("Outer: " + i);  
        for (int j = 1; j <= 3; j++) {  
            System.out.println(" Inner: " + j);  
        }  
    }  
}
```


NESTED FOR LOOP SYNTAX IN ITERATING 2D ARRAY

```
for (int row =0; row<identifier.length;row++){  
    for (int col =0; col<identifier[row].length; col++){  
        System.out.println(identifier[row][col]);  
    }  
}
```

ITERATING 2D ARRAY USING NESTED FOR LOOP

```
public class Array {  
  
    public static void main(String[] args) {  
  
        String [][] User_Info = {{ "Kate", "Mark", "Cris"}, {"Kate123", "Mark345", "Crisabc"}};  
  
        for (int row = 0; row < User_Info.length; row++) {  
  
            for (int col = 0; col < User_Info[row].length; col++) {  
  
                System.out.println(User_Info[row][col]);  
  
            }  
  
            System.out.println();  
  
        }  
  
    }  
}
```

Output - Array (run)



run:

Kate

Mark

Cris

Kate123

Mark345

Crisabc

BUILD SUCCESSFUL (total time: 0 seconds)

FOR EACH LOOP IN ITERATING 2D ARRAY

Are used to iterate through collections like arrays easily without knowing the size of the collection.

SYNTAX:

```
For(datatype identifier:collection) {  
  //to do code  
}
```

FOR EACH LOOP ITERATING ARRAY

```
public class Array {  
  
    public static void main(String[] args) {  
  
        String [] User_Info = {"Kate", "Mark", "Cris"};  
  
        for(String user:User_Info){  
            System.out.println(user);  
        }  
    }  
}
```

Output - Array (run)



run:



Kate



Mark



Cris



BUILD SUCCESSFUL (total time: 0 seconds)

FOR EACH LOOP ITERATING ARRAY

```
public class Array {  
  
    public static void main(String[] args) {  
  
        int [] numbers ={1, 2,3,4,5,5};  
        for(int num:numbers){  
            System.out.println(num);  
        }  
    }  
}
```

NESTED FOR EACH LOOP SYNTAX IN ITERATING 2D ARRAY

For(String [] Var_name :Array_identifer){ - holds the array

For(String NewVar_name: Var_name){ - holds the value of each
array

}

}

IN ITERATING 2D ARRAY USING NESTED FOR EACH LOOP

```
public class Array {  
  
    public static void main(String[] args) {  
  
        String [][] User_Info = {"Kate", "Mark", "Cris"}, {"Katel23", "Mark345", "Crisabc"};   
  
        for(String user[]:User_Info){  
            for(String infos:user){  
                System.out.println(infos);  
            }  
            System.out.println();  
        }  
    }  
}
```

Output - Array (run)

run:
Kate
Mark
Cris

Katel23
Mark345
Crisabc

BUILD SUCCESSFUL (total time: 0 seconds)